

Dark energy

The universe has it too; not just the Sith Lords, it apparently accounts for 74% of all the matter in the universe

Hubble revived

After a decade of faithful service, the hubble telescope died. It's back though, till 2010, thanks to some desperate repair work by NASA

Spaceyard

The Russian and European space agencies are planning to build a spaceship yard for future interplanetary travel

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Einstein died an annoyed man. After he shot to international stardom with his theory of General Relativity, he had devoted his life to a “unified theory” that connected the two major forces in the universe – gravity and electromagnetism – but it only ended in frustration. But in the sixties, scientists came across a mathematical equation that promised to change everything we knew. String Theory says that everything – from quark to atom to human to planet – is fundamentally composed of little vibrating strings.

You're probably asking, “So what?”

The reason string theory is such a big deal is that if true, it successfully describes all four natural forces – gravity, electromagnetism, and strong and weak nuclear forces – with a mathematical elegance that other theories only

dream about. But before we understand string theory (and that's a relative term), we must understand why it needs to exist.

The story of gravity

When Newton described gravity, it made perfect mathematical sense. More importantly, it could be verified experimentally – scientists love it when that happens. Even so, Newton and his fellows couldn't figure out how gravity really worked. Do bodies just reach out and grab each other with invisible arms?

Enter Einstein. He saw the four dimensions – length, width, depth and time – as a fabric, capable of stretching and warping under influence. This influence, he said, came from the planets and stars. Imagine a large piece of cloth, stretched to firmness. If you were to toss a watermelon on it, it would create a sizable depression on the cloth. If you threw a tennis ball into the mix, it would not only create a smaller depression, it would start to roll

towards the watermelon. This is essentially how gravity works – stars and planets create their own depressions – called gravity wells – in space-time, and small objects either fall into these wells, or swirl around them as satellites.

While the world was still busy praising this new theory, Einstein wanted to take it one step further. Only a few years earlier, Maxwell Planck had proved that electricity and magnetism were just manifestations of the same force – electromagnetism. To Einstein, it seemed logical that electromagnetism and gravity were manifestations of something bigger, and set out to find out what that was. The problem, however, was that electromagnetism and gravity are vastly different.

Space oddity

We tend to think of gravity as this strong, anchoring force that's holding the universe together – because, after all, without gravity, everything will just float into nothingness.

In reality, though, gravity is one of the weakest forces in the universe. It may be strong enough to keep you in your chair, but it's too weak to pull you through the chair, towards the ground, to the center of the Earth. Or to prevent you from jumping, for that matter.

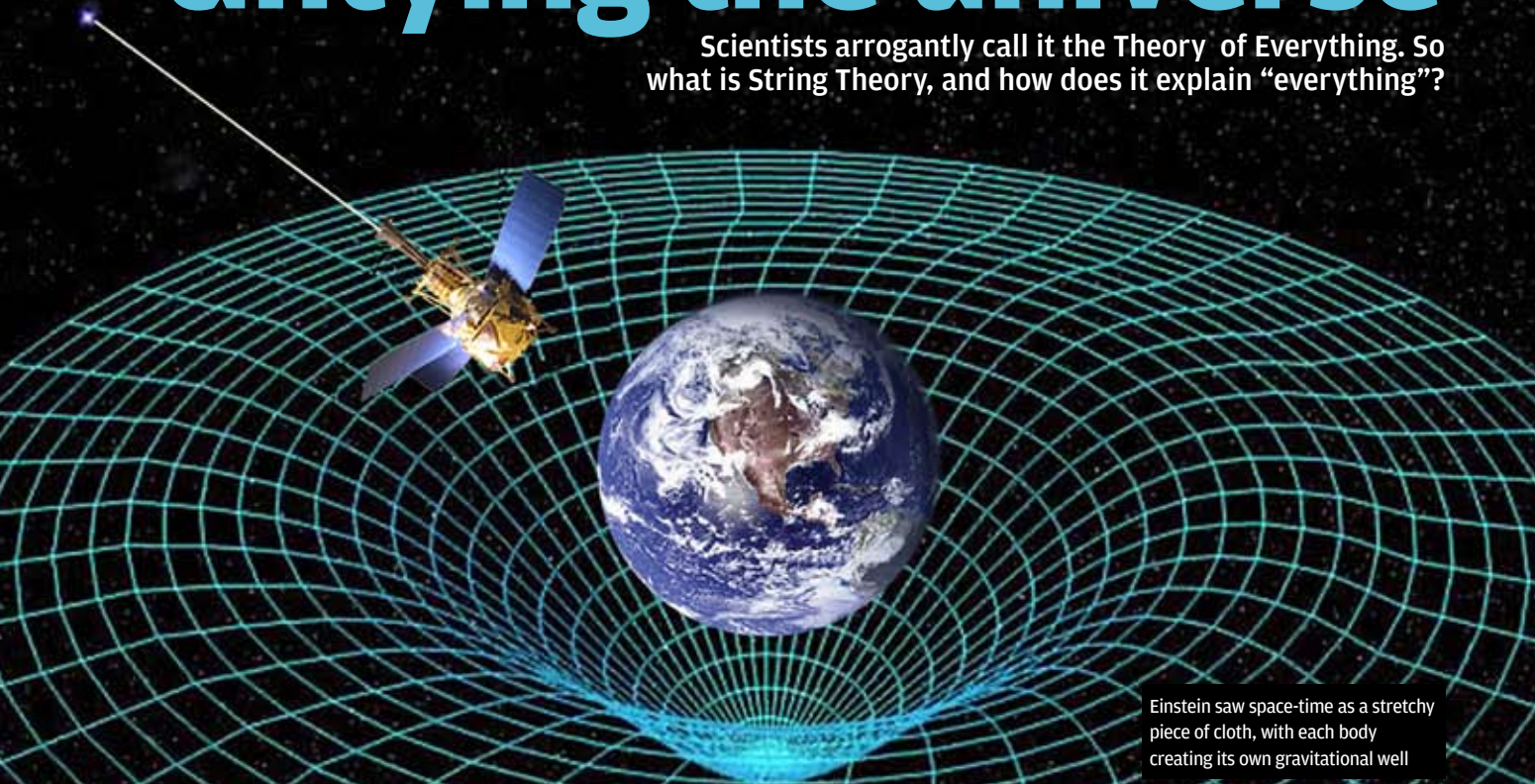
Let's understand this better.

You and your chair (and everything else) are made of atoms – protons and neutrons at the centre, with electrons orbiting around them. This outer shell of electrons is negatively charged, so when the atoms of your body come close to those of your chair, they repel each other. This repulsion – an electric force – is so strong that even if you turned the tallest, pointiest building in the world upside-down and made it cover only a square foot of ground, it'll never go through it. In fact, at the atomic level, gravity is so weak that scientists often disregard it in their calculations.

If gravity and electromagnetism come from the same place, it doesn't make

Untying the universe

Scientists arrogantly call it the Theory of Everything. So what is String Theory, and how does it explain “everything”?



Einstein saw space-time as a stretchy piece of cloth, with each body creating its own gravitational well